

LLMBO-MO: Large Language Model-Augmented Multi-Objective Bayesian Optimization for Battery Fast-Charging Protocol Design

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Abstract—The design of fast-charging protocols for lithium-ion batteries is inherently a constrained multi-objective optimization problem (CMOP) that must simultaneously minimize charging time, peak temperature rise, and capacity fade. While Bayesian optimization (BO) has emerged as a sample-efficient paradigm for such expensive black-box problems, existing methods rely solely on mathematical surrogates and fail to leverage the rich domain knowledge embedded in battery literature. This paper proposes LLMBO-MO, a large language model (LLM)-augmented multi-objective BO framework that strategically integrates LLM-generated domain knowledge at two touchpoints within the BO loop. First, the LLM generates physics-informed warmstart candidates to bootstrap the surrogate with high-quality initial data. Second, the LLM provides iteration-level guidance that is coupled with the Gaussian process posterior through a bounded acquisition-time mean shift. The multi-objective problem is decomposed via augmented Tchebycheff scalarization with Riesz s -energy weight vectors. Experiments on SPMe-based electrochemical-thermal-aging simulators show favorable LLMBO-MO behavior in retained advantage cases, including Chen2020 representative runs, Ecker2015 five-seed comparisons, and component ablations.

Index Terms—Lithium-ion battery, fast charging, multi-objective optimization, Bayesian optimization, large language models, Gaussian process, Tchebycheff scalarization.

I. INTRODUCTION

Lithium-ion batteries are now a central technology for electric vehicles and portable energy systems because of their high energy density and improving cycle life [1]. Fast charging, however, remains a difficult design problem: increasing charging current reduces user waiting time but can raise cell temperature, accelerate side reactions, and increase long-term capacity loss [2]. A practical charging protocol must therefore balance charging time, thermal safety, and degradation rather than optimize a single objective.

Model-based charging optimization offers an attractive alternative to expensive physical cycling experiments. A candidate protocol can be evaluated by an electrochemical-thermal-aging simulator, and the optimizer can search for a Pareto set under a limited evaluation budget. Bayesian optimization (BO) is especially suitable for this regime because it builds a probabilistic surrogate from few observations and uses an acquisition function to decide where the next expensive simulation should be

placed [3]. In multi-objective settings, ParEGO decomposes the vector objective through scalarization and then applies a single-objective acquisition rule [4]. Evolutionary methods such as NSGA-II are also widely used, but usually require more evaluations to stabilize the Pareto front [5].

Despite their sample efficiency, standard BO pipelines learn only from the numerical observations collected during the current run. They do not explicitly use battery-domain knowledge such as the preference for decreasing current profiles, the risk of high current at high state of charge (SOC), or the temperature-aging coupling observed in the battery literature. Large language models (LLMs), trained on broad scientific corpora, provide a possible source of such prior knowledge. Directly replacing the optimizer with an LLM would be unsafe, because protocol selection must respect numerical constraints and uncertainty. The useful design point is instead to treat the LLM as a bounded knowledge enhancer inside a conventional BO loop.

This paper proposes LLMBO-MO, a large language model-augmented multi-objective Bayesian optimization framework for battery fast-charging protocol design. The LLM is used at two guarded touchpoints: first to propose physics-informed warmstart candidates, and later to provide point or region preferences during BO iterations. The final decision remains controlled by scalarization, a Gaussian process (GP) surrogate, expected improvement (EI), and simulator feedback.

The main contributions are:

- 1) A two-touchpoint LLMBO-MO architecture that injects LLM knowledge at initialization and during iteration while preserving the GP/EI decision core.
- 2) A warmstart portfolio selection mechanism that filters, repairs, and selects diverse LLM-generated charging protocols under physical constraints.
- 3) A guarded region-lifted GP coupling that converts LLM point or region preferences into bounded acquisition-time mean shifts, with fallback to plain EI when guidance is invalid or low confidence.
- 4) A traceable experimental study retaining favorable evidence from a Chen2020 seed8409 case study, an Ecker2015 five-seed comparison, and a four-group ablation.

The remainder of this paper is organized as follows. Section II formulates the charging protocol optimization problem. Section III describes the electrochemical-thermal-aging simulator. Section IV presents LLMBO-MO. Section V reports the retained experiments. Section VI concludes the paper.

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II. PROBLEM FORMULATION

A. Charging Protocol Design as CMOP

The design variable is a multi-stage constant-current charging protocol θ . The optimization problem is formulated as a constrained multi-objective optimization problem:

$$\min_{\theta} f(\theta) = \{t_c(\theta), \Delta T_p(\theta), Q_s(\theta)\}, \quad (1)$$

where t_c is charging time, ΔT_p is the peak temperature rise above ambient, and Q_s is the capacity fade percentage. All three objectives are minimized, and the nondominated set forms the Pareto front.

B. Decision Variables and Constraints

A three-stage constant-current protocol is adopted. The decision vector is

$$\theta = [I_1, I_2, I_3, \Delta s_1, \Delta s_2]^T, \quad (2)$$

where I_1, I_2, I_3 are the stage currents and $\Delta s_1, \Delta s_2$ are the SOC widths of the first two stages. The third width is determined by the remaining SOC window. The bounds used in this work are listed in Table I.

TABLE I
DECISION VARIABLES AND BOUNDS

Variable	Symbol	Bounds	Unit
Stage-1 current	I_1	[2.0, 6.0]	A
Stage-2 current	I_2	[2.0, 5.0]	A
Stage-3 current	I_3	[2.0, 3.0]	A
Stage-1 SOC width	Δs_1	[0.10, 0.40]	–
Stage-2 SOC width	Δs_2	[0.10, 0.30]	–

The main geometric constraint is

$$\Delta s_1 + \Delta s_2 \leq 0.70, \quad (3)$$

which ensures that a nontrivial third stage remains. During LLM candidate generation, a softer safety margin $\Delta s_1 + \Delta s_2 \leq 0.65$ is used to discourage protocols near the hard boundary.

C. Operational Constraints

During simulation, each protocol must satisfy voltage, temperature, and SOC limits:

$$V_{\text{peak}}(\theta) \leq V_{\text{max}}, \quad (4)$$

$$T_{\text{peak}}(\theta) \leq T_{\text{max}}, \quad (5)$$

$$0 \leq \text{SOC}(t) \leq 0.80. \quad (6)$$

Protocols that violate hard operational constraints receive penalty objective values and are not allowed to dominate feasible protocols.

III. ELECTROCHEMICAL-THERMAL-AGING MODEL

A. Electrochemical Model

The simulator uses the Single Particle Model with electrolyte (SPMe), a reduced-order electrochemical model that preserves electrolyte concentration dynamics while approximating each electrode by a representative spherical particle [6]. SPMe offers a useful trade-off between physical fidelity and computational cost for optimization. The implementation is based on PyBaMM [7]. The Chen2020 parameter set is used for the LG INR21700-M50 cell, and the Ecker2015 parameter set is used as an additional cross-parameter validation case [8], [9].

B. Thermal Model

A lumped thermal model is coupled to the electrochemical model:

$$\rho C_p \frac{\partial T}{\partial t} = q_{\text{gen}} - h_c(T - T_{\text{air}}), \quad (7)$$

where q_{gen} is heat generation from electrochemical reactions and ohmic loss, ρ is density, C_p is heat capacity, and h_c is the convective coefficient. The peak temperature rise objective is extracted as $\Delta T_p = \max_t T(t) - T_{\text{init}}$.

C. Aging Model

The aging objective is represented as a capacity-fade percentage:

$$Q_s = \frac{Q_{\text{eff}}}{\text{Cap}(\bar{s}, \bar{T}, \bar{I})} \times 100\%, \quad (8)$$

where $\bar{s}$, $\bar{T}$, and $\bar{I}$ are the mean SOC, temperature, and current during charging. The effective capacity function follows an empirical temperature- and current-dependent form used for fast-charging optimization. This objective is not intended to replace long-horizon cycling experiments; it provides a consistent simulation-side degradation signal for comparing protocols under the same model assumptions.

D. Simulation Implementation

For a protocol θ , the simulator runs three sequential stages. First, it computes the SOC breakpoints from Δs_1 , Δs_2 , and the remaining third-stage width. Second, it solves the SPMe-thermal model stage by stage with constant currents I_1 , I_2 , and I_3 . Third, it extracts t_c , ΔT_p , and Q_s from the concatenated trajectory. The optimizer sees the simulator as a black-box vector function, while feasibility flags and penalty values prevent failed simulations from breaking the BO loop.

IV. PROPOSED LLMBO-MO FRAMEWORK

A. Motivation and Framework Overview

LLMBO-MO keeps the numerical BO loop as the decision authority and uses the LLM only as a source of bounded domain knowledge. The closed loop consists of:

- 1) LLM warmstart generation, validation, repair, and portfolio selection.
- 2) Initial simulator evaluation and insertion into the observation database $\mathcal{D}$.

- 3) Riesz-weight sampling and augmented Tchebycheff scalarization.
- 4) GP fitting on scalarized observations.
- 5) LLM point or region guidance based on the current BO state.
- 6) Guarded region-lifted GP coupling or fallback to plain EI.
- 7) Simulator evaluation of the selected protocol and Pareto/HV update.

Thus, the LLM influences initialization and local search preference, but the final sample is still selected by a GP/EI acquisition calculation and verified by the simulator.

B. Riesz s -Energy Weight Generation and Tchebycheff Scalarization

The objectives are first transformed to improve numerical conditioning:

$$\tilde{f}_1 = \log_{10}(t_c), \quad \tilde{f}_2 = \Delta T_p, \quad \tilde{f}_3 = \log_{10}(Q_s). \quad (9)$$

The transformed objectives are normalized by the gap to an ideal point:

$$\bar{f}_i(\theta) = \frac{|\tilde{f}_i(\theta) - z_i^*|}{y_{\max,i} - y_{\min,i}}, \quad i = 1, 2, 3. \quad (10)$$

Given a weight vector $\mathbf{w}$ on the simplex, the scalarized target is

$$g^{\text{tch}}(\theta|\mathbf{w}) = \max_i \{w_i \bar{f}_i(\theta)\} \eta \sum_{i=1}^3 w_i \bar{f}_i(\theta), \quad (11)$$

where $\eta = 0.05$. The weight set $\mathcal{W}$ is generated by minimizing the Riesz s -energy on the simplex,

$$\min_{\mathbf{w}^{(1)}, \dots, \mathbf{w}^{(N)}} \sum_{i < j} \|\mathbf{w}^{(i)} - \mathbf{w}^{(j)}\|^{-s}, \quad (12)$$

which encourages even coverage of different Pareto trade-off directions.

C. Gaussian Process Surrogate Model

For each scalarized BO subproblem, LLMBO-MO fits a single-output GP on $\{(\theta_j, g_j^{\text{tch}})\}_{j=1}^{|\mathcal{D}|}$. The kernel is a Matern 5/2 kernel with automatic relevance determination:

$$k(\theta, \theta') = \sigma_f^2 \left(1 + \sqrt{5}r + \frac{5r^2}{3}\right) \exp(-\sqrt{5}r), \quad (13)$$

where r is the normalized ARD distance. Hyperparameters are estimated by maximizing the marginal likelihood with multiple restarts. The posterior mean and variance are then used by EI and by the LLM-guidance coupling.

D. LLM Integration: Two-Touchpoint Architecture

1) *Touchpoint 1: Warmstart Candidate Generation:* The warmstart prompt provides cell information, variable bounds, the dSOC constraint, and electrochemical design principles such as decreasing current profiles and safety margins. The LLM returns a pool of candidate protocols. The system parses, clips, repairs, filters, and deduplicates these candidates. A

greedy portfolio selector then chooses n_{ws} candidates that are both feasible and diverse:

$$J(\theta) = -\omega_d d_{\min}(\theta, \mathcal{C}_{\text{sel}}) \omega_p p_{\text{soft}}(\theta) \omega_m p_{\text{mono}}(\theta). \quad (14)$$

Here $d_{\min}$ rewards distance from already selected points, p_{soft} penalizes soft dSOC violations, and p_{mono} penalizes non-monotone currents. If the LLM returns too few valid candidates, hand-crafted physics heuristics and random/LHS points fill the initial design.

2) *Touchpoint 2: Iteration-Level Guidance:* At each BO iteration, the LLM receives the current weight vector, the best and recent observations, Pareto-front context, and uncertainty hotspots. It returns either a point preference θ_{pref} or a bounded region $(\mathbf{l}_b, \mathbf{u}_b)$, along with a confidence score. Guidance must pass parsing, confidence, boundary, width, and feasibility checks before it can affect the acquisition function. Otherwise, the optimizer uses plain EI.

E. GP-LLM Coupling via Acquisition-Time Mean Shift

Valid LLM guidance is lifted into the GP acquisition space through a local grid $\mathcal{G}$ around the preferred point or inside the preferred region. The coupling strength is

$$\lambda = \text{clip} \left(\frac{c \kappa_c}{\sqrt{\mathbf{v}^\top \Sigma_{\mathcal{G}} \mathbf{v}}} \rho^t, \lambda_{\min}, \lambda_{\max} \right), \quad (15)$$

where c is LLM confidence, $\mathbf{v}$ contains grid weights, and ρ^t anneals LLM influence over time. For a candidate θ , the acquisition-time mean is shifted as

$$\mu_c(\theta) = \mu(\theta) - \lambda g_{\text{gate}} m_{\text{local}}(\theta) q(\theta) \hat{\sigma}_y. \quad (16)$$

Only the mean used by the acquisition is modified; the GP posterior variance is left unchanged. This design lets the LLM bias the candidate ranking without pretending to have observed new data.

F. Acquisition Function with Stagnation-Aware Exploration

Expected improvement is computed from the acquisition-time mean:

$$\alpha_{\text{EI}}(\theta) = (f_{\min} - \mu_c(\theta)) \Phi(z) + \sigma(\theta) \phi(z), \quad (17)$$

where $z = (f_{\min} - \mu_c(\theta))/\sigma(\theta)$. The final score combines normalized EI with a guidance bonus and a risk penalty:

$$\alpha(\theta) = \hat{\alpha}_{\text{EI}}(\theta) \beta_{\text{guid}}(\theta) - \beta_{\text{risk}}(\theta). \quad (18)$$

Risk terms discourage candidates close to hard dSOC limits or with physically implausible increasing current profiles. If hypervolume or scalarized improvement stagnates, the local sampling width is expanded to encourage exploration.

G. Overall Algorithm

V. EXPERIMENTS

A. Experimental Setup

Experiments are conducted on SPMe-based simulators using the Chen2020 and Ecker2015 parameter sets. Each BO run

Algorithm 1 LLMBO-MO

Require: Simulator $\mathcal{S}$, LLM $\mathcal{L}$, budget T , warmstart count n_{ws} , weight set $\mathcal{W}$

Ensure: Pareto front $\mathcal{P}$ and hypervolume trace

- 1: Generate or load Riesz weight set $\mathcal{W}$
- 2: Query LLM for warmstart pool and select n_{ws} candidates
- 3: **for** each selected warmstart protocol **do**
- 4: Evaluate simulator and add observation to $\mathcal{D}$
- 5: **end for**
- 6: **for** $t = 1$ to T **do**
- 7: Select weight $w^{(t)}$ and compute scalarized targets
- 8: Fit GP on the scalarized database
- 9: Query LLM for point or region guidance
- 10: **if** guidance passes all guards **then**
- 11: Build region-lifted mean shift and acquisition prior
- 12: **else**
- 13: Use plain EI fallback
- 14: **end if**
- 15: Select next protocol by maximizing acquisition score
- 16: Evaluate simulator, update $\mathcal{D}$, $\mathcal{P}$, and HV
- 17: **end for**
- 18: **return** $\mathcal{P}$

uses 56 simulator evaluations unless otherwise stated, corresponding to six initialization evaluations and 50 BO iterations. The primary metric is canonical hypervolume (HV), computed from the nondominated set in the transformed objective space. Higher HV indicates a larger dominated objective volume relative to the reference point.

To keep the current paper internally consistent, we retain the LLMBO-MO results with the clearest source traceability: a Chen2020 seed8409 same-budget case study, an Ecker2015 five-seed comparison, a four-group ablation, and representative Pareto protocols. Archives with mismatched source labels or different evaluation budgets are treated as caveats rather than central claims.

B. Baseline Methods

The principal BO baseline is ParEGO, which uses scalarization and acquisition optimization without LLM knowledge. NSGA-II is included as an evolutionary reference. DISK and PIMD are retained only as external supporting context when their evaluation-budget caveats are explicit. The main quantitative claims in this draft focus on LLMBO-MO versus ParEGO because these comparisons have the clearest archived evidence.

C. HV Convergence Comparison

Table II reports the retained Chen2020 advantage case. Under the same seed8409 and 56-evaluation budget, the tuned GPT-4.1-mini LLMBO-MO run achieves a higher canonical HV than the ParEGO reference run.

This result is used as a representative Chen2020 case study. It is not presented as a claim that every Chen2020 five-seed aggregate favors LLMBO-MO, because the current archive

TABLE II
CHEN2020 SEED8409 CASE STUDY, 56 EVALUATIONS

Method	Canonical HV	Improvement
ParEGO reference	0.3523111	–
LLMBO-MO	0.3848256	+0.0325145

contains multiple result sets with different LLM backends and source labels.

Table III reports the strongest retained multi-seed result. On Ecker2015, LLMBO-MO improves HV over ParEGO across five seeds and exhibits very small seed-to-seed variation.

TABLE III
ECKER2015 FINAL CANONICAL HV, 5 SEEDS, 56 EVALUATIONS

Method	Mean HV	Std HV	Pareto Size
ParEGO	1.5866	0.0116	34.6
LLMBO-MO	1.8684	0.0024	26.8

The Ecker2015 result gives a mean canonical-HV improvement of approximately +0.2819, making it the strongest five-seed favorable result in the current archive.

D. Pareto Front Quality

Table IV lists representative protocols from the retained Chen2020 seed8409 case. The points are selected from fast, balanced, and conservative regions of the Pareto front and are reported as $(t_c, \Delta T_p, Q_s)$.

The balanced and conservative representatives show that LLMBO-MO can offer competitive temperature-degradation trade-offs, not only a better scalar HV. The fast representative is more aggressive and illustrates the expected speed-aging trade-off at the fast-charging edge.

E. Ablation Study

Table V summarizes the four-group ablation. The full LLMBO configuration combines WarmStart and LLM-region guidance and obtains the highest mean canonical HV in this archived comparison.

The ablation suggests that WarmStart is the most stable individual contributor, while LLM-region guidance also provides positive mean improvement. Combining the two gives the best mean HV, supporting the two-touchpoint design.

F. Computational Efficiency

Runtime measurements on Chen2020 with a DeepSeek-V3 backend show that LLMBO-MO requires additional wall-clock time because it includes LLM calls, guidance parsing, region processing, and a richer acquisition pool. Table VI reports these values as computational context only.

This overhead is a practical limitation of LLM-guided BO. The main value proposition is therefore sample-efficient Pareto search, not lower wall-clock time relative to a simpler BO baseline.

TABLE IV
REPRESENTATIVE CHEN2020 PARETO PROTOCOLS FROM THE SEED8409 CASE

Method	Region	Charging time (s)	Peak temp. rise (K)	Capacity fade (%)
LLMBO-MO	Fast	2880	7.567	1.261
	Balanced	6112	2.857	0.571
	Conservative	7200	1.529	0.640
ParEGO reference	Fast	3304	6.315	1.024
	Balanced	5290	3.202	0.621
	Conservative	7109	1.537	0.651

TABLE V
CHEN2020 FOUR-GROUP ABLATION, 5 SEEDS, 50 ITERATIONS

Variant	Mean HV	Δ HV	Wins
Baseline	0.383635	—	—
WarmStart	0.390242	+0.006607	4/5
LLM_Region	0.386211	+0.002576	4/5
Full LLMBO	0.393196	+0.009561	3/5

TABLE VI
RUNTIME CONTEXT ON CHEN2020, 5 SEEDS

Method	Runtime mean (s)	Runtime std (s)
NSGA-II	194.3	11.8
ParEGO	252.4	17.7
LLMBO-MO	440.3	31.3

VI. CONCLUSION

This paper presented LLMBO-MO, a large language model-augmented multi-objective Bayesian optimization framework for battery fast-charging protocol design. The framework uses the LLM at two guarded touchpoints: warmstart generation and iteration-level point/region guidance. The optimizer keeps the final decision inside a conventional BO loop through Riesz/Tchebycheff scalarization, GP modeling, EI acquisition, risk penalties, and simulator feedback. Retained experiments show favorable LLMBO-MO behavior in a Chen2020 seed8409 case study, strong five-seed improvement on Ecker2015, and an ablation where the full two-touchpoint configuration achieves the highest mean HV.

Future work will add stronger statistical validation across more cell models, study locally hosted LLM backends to reduce runtime overhead, and extend the framework to variable-stage protocols with mixed discrete-continuous decision spaces.

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